

⚡ · Inference

Bayesian inference gives us $p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$. We can evaluate this for any θ . But everything we want—means, variances, credible intervals—requires integration over the posterior, and these integrals are intractable in high dimensions.

This chapter covers two complementary approaches: **sampling methods** (MCMC) that give exact samples but can be slow, and **variational inference** that gives fast approximations. Understanding both—and their tradeoffs—is essential.

1 Intractable integrals

The difficulty is twofold. First, the number of evaluations explodes: grid-based quadrature with just 10 points per dimension requires 10^d evaluations, which for $d = 15$ (typical for gravitational wave parameter estimation) means 10^{15} —impossible.

Second, the posterior concentrates in a tiny region. If the posterior constrains each parameter to 10% of the prior range, the high-probability region occupies a fraction $(0.1)^{15} \approx 10^{-15}$ of the total volume. Almost all grid points fall where $p(\theta \mid x) \approx 0$.

Monte Carlo integration provides a way forward. Given samples $\{\theta_i\}$ from the posterior, any integral becomes an average:

$$\int f(\theta) p(\theta \mid x) d\theta \approx \frac{1}{N} \sum_{i=1}^N f(\theta_i) \quad (1.1)$$

The error scales as $1/\sqrt{N}$, independent of dimension. This is the key insight: **sampling is for doing integrals**. If we can generate samples from the posterior, we can compute any expectation we want.

The question becomes: how do we generate samples from a distribution we can only evaluate up to a constant?

2 The Metropolis–Hastings algorithm

Imagine exploring a landscape where elevation represents probability—higher means more probable. You want to collect samples that represent this landscape (spending more time in high-probability regions), but you can only check elevation where you’re standing.

Metropolis–Hastings is a random walk with a clever acceptance rule:

1. **Propose:** From current position θ , suggest a move to θ' drawn from some proposal distribution $q(\theta' \mid \theta)$ —often a Gaussian centered at the current position.
2. **Accept or reject:** Compute the acceptance probability:

$$\alpha = \min \left(1, \frac{p(\theta' \mid x)}{p(\theta \mid x)} \right) \quad (2.1)$$

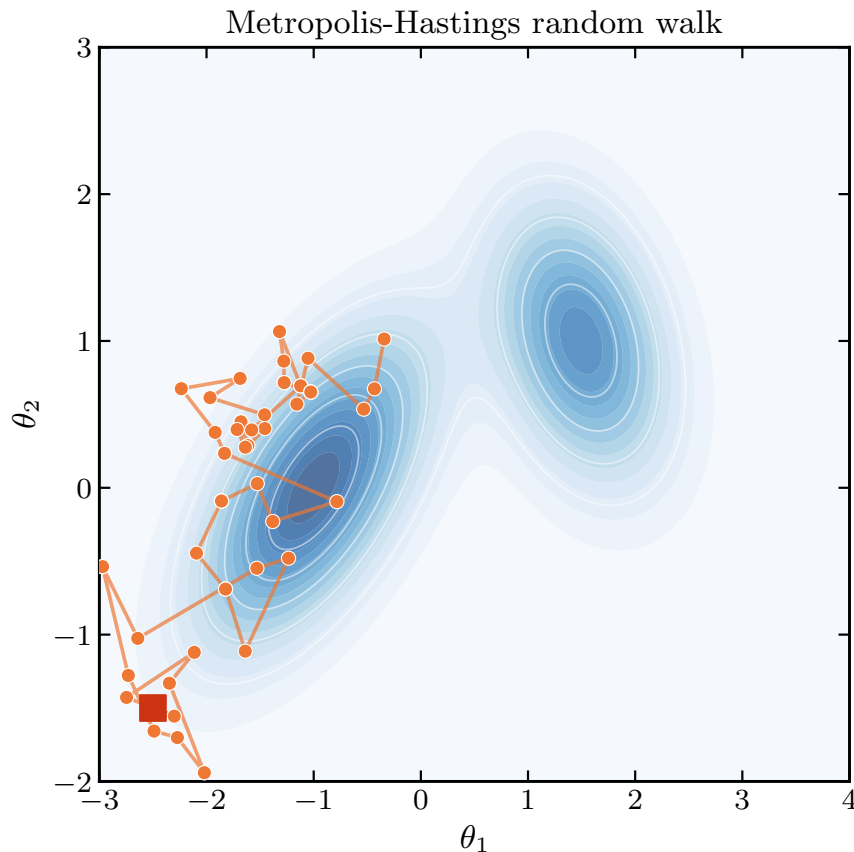


Figure 1. MCMC as landscape exploration. The surface represents a posterior distribution over two parameters. The algorithm proposes moves (arrows), accepts those that go uphill or pass a random test when going downhill, and traces out a path that spends more time in high-probability regions.

If the new spot is higher probability: always accept. If lower: accept with probability α .

3. **Record:** If accepted, move to θ' ; otherwise stay at θ . Either way, record the current position as a sample.

Accepting “downhill” moves with probability proportional to how much lower they are prevents getting stuck on one peak while still spending more time in high-probability regions.

Why normalizing constants cancel. The acceptance ratio only involves $p(\theta' | x)/p(\theta | x)$. Since both share the same normalizing constant $p(x)$, it cancels:

$$\frac{p(\theta' | x)}{p(\theta | x)} = \frac{p(x | \theta') p(\theta')}{p(x | \theta) p(\theta)} \quad (2.2)$$

We only need the unnormalized posterior—likelihood times prior.

3 Tuning and convergence

The proposal distribution has a free parameter: step size σ . The choice matters enormously.

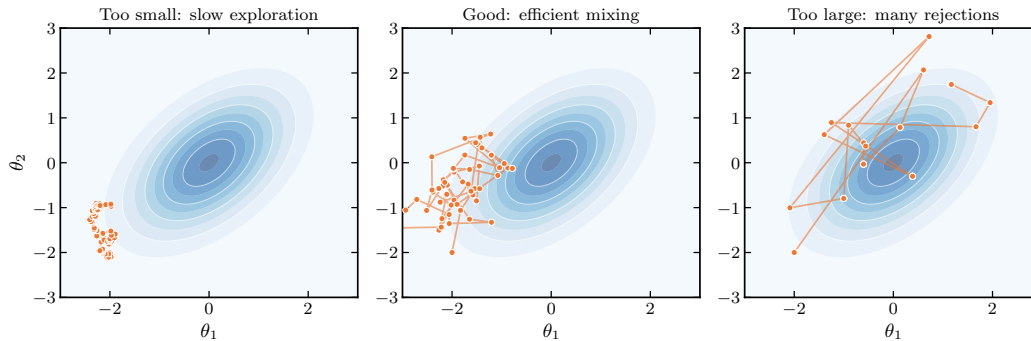


Figure 2. The step size tradeoff. (a) Step size too small: nearly all proposals are accepted, but exploration is slow—the chain takes many steps to traverse the distribution. (b) Step size too large: most proposals land in low-probability regions and are rejected—the chain gets stuck. (c) Step size well-tuned: efficient exploration through balanced acceptance.

- **Too small:** Nearly all proposals are accepted, but exploration is glacially slow.
- **Too large:** Most proposals land in low-probability regions and are rejected. The chain stays stuck.

A good acceptance rate is roughly 25% in high dimensions—a balance between moving and accepting.

Autocorrelation. MCMC samples are correlated—each is near the previous. The **autocorrelation time** τ measures how many steps until samples become approximately independent. If $\tau = 100$, then every 100th sample is effectively independent; the effective sample size is $N_{\text{eff}} = N/\tau$. Autocorrelation time is the most important diagnostic: it tells you how many samples you actually have.

4 The failure of random walks in high dimensions

Random walk Metropolis scales poorly with dimension. The autocorrelation time grows as $\tau \propto d$ —in d dimensions, you need d times more samples.

Why does this happen? In high dimensions, probability concentrates in a thin shell called the *typical set*. For a d -dimensional Gaussian, this shell lies at radius $r \sim \sqrt{d}$ from the mode, with thickness scaling as $1/\sqrt{d}$.

A random walker on this shell faces a geometric problem: most directions are perpendicular to the shell. Only a tiny fraction of moves stay along the shell and make progress. As dimension increases, this fraction shrinks.

5 Hamiltonian Monte Carlo

The posterior has structure—the gradient $\nabla \log p(\theta | x)$ tells us which way is “uphill.” Random walk Metropolis ignores this. Hamiltonian Monte Carlo (HMC) uses the gradient to make informed, directed moves.

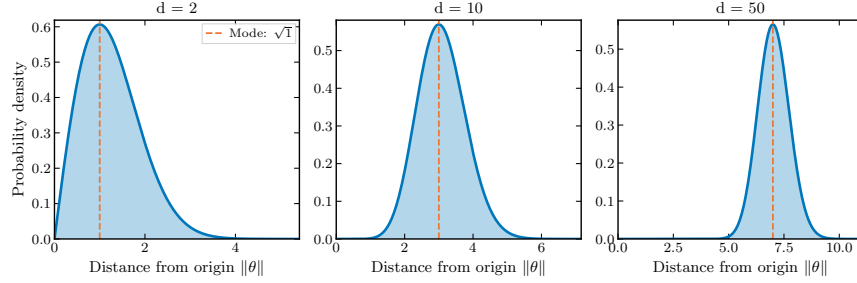


Figure 3. The typical set in high dimensions. For a d -dimensional Gaussian, most probability mass lies not at the mode (where density is highest) but in a thin shell at radius $\sqrt{d}$. Volume grows rapidly with radius, overwhelming the density decay. In $d = 100$ dimensions, the mode is far from the typical set.

5.1 The physics picture

Imagine a ball rolling on a surface where height is negative log-probability. A ball given a push rolls through valleys, slows climbing hills, turns around, and oscillates—naturally spending time in low-energy (high-probability) regions.

Augment parameters θ with momentum p . The Hamiltonian (total energy) is:

$$H(\theta, p) = \underbrace{-\log p(\theta | x)}_{\text{potential energy}} + \underbrace{\frac{1}{2} p^\top p}_{\text{kinetic energy}} \quad (5.1)$$

Hamilton's equations give the dynamics:

$$\frac{d\theta}{dt} = p \quad (\text{position changes according to momentum}) \quad (5.2)$$

$$\frac{dp}{dt} = \nabla \log p(\theta | x) \quad (\text{momentum changes according to gradient}) \quad (5.3)$$

5.2 The algorithm

1. **Sample momentum:** Draw $p \sim \mathcal{N}(0, I)$ —give the ball a random push.
2. **Simulate dynamics:** Integrate Hamilton's equations for time T using a leapfrog integrator.
3. **Metropolis correction:** Accept with probability $\min(1, e^{-H(\theta', p') + H(\theta, p)})$.

The payoff. HMC achieves autocorrelation time τ nearly independent of dimension—a massive improvement over random walk ($\tau \propto d$). The cost is requiring gradients, but automatic differentiation makes this routine.

6 Diagnostics

We can never prove MCMC has converged, but failures can often be detected.

Trace plots. Plot each parameter against iteration number. A converged chain looks like stationary noise—fluctuating around a stable mean with no trends.

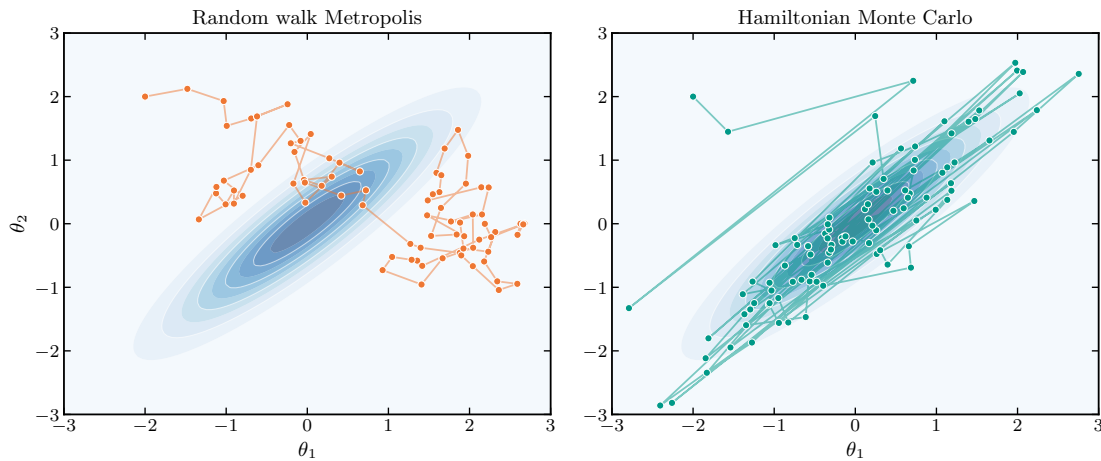


Figure 4. Random walk vs. Hamiltonian Monte Carlo. (a) Random walk Metropolis: the chain diffuses slowly, taking many small steps to traverse the distribution. (b) HMC: gradient-informed trajectories make long, directed moves across the posterior, exploring efficiently even in high dimensions.

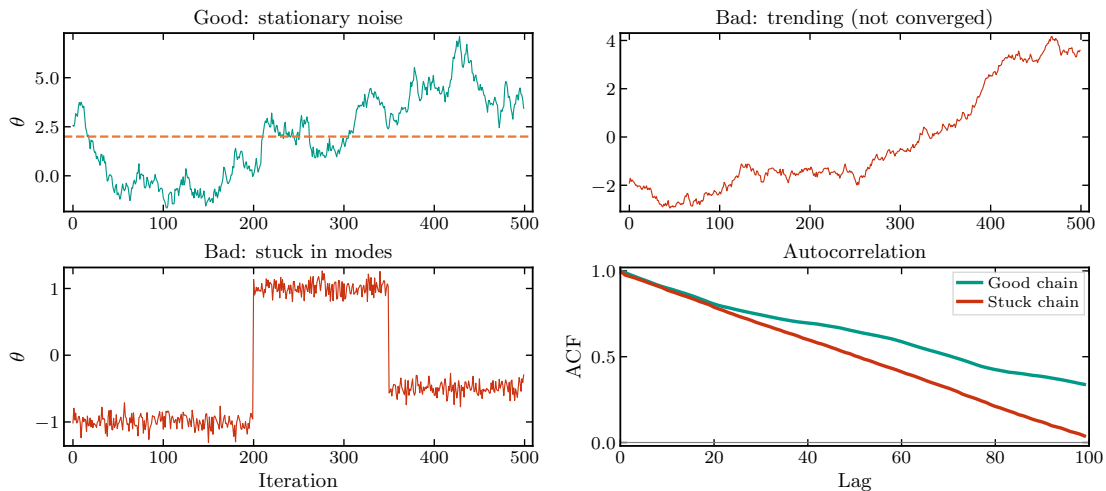


Figure 5. MCMC diagnostics via trace plots. (a) Good mixing: the chain fluctuates around a stable mean with no trends, indicating convergence. (b) Poor mixing: the chain shows slow drift or gets stuck in regions, suggesting more samples are needed or the sampler needs tuning.

Multiple chains. Run several chains from different starting points. If they converge to the same distribution, that's evidence of convergence. The $\hat{R}$ statistic compares within-chain variance to between-chain variance; $\hat{R} \approx 1$ indicates agreement.

Effective sample size. The effective sample size is $N_{\text{eff}} = N/\tau$: from N correlated samples, we have information equivalent to N_{eff} independent ones. This is what you should report, not the raw number of samples.

7 Using samples

Once we have posterior samples, we can answer any question about the posterior.

Point estimates. Posterior mean: $\bar{\theta} = \frac{1}{N} \sum_i \theta_i$. Do not report the highest-probability sample—MCMC is a sampler, not an optimizer.

Credible intervals. The 16th and 84th percentiles give 68% credible intervals; 2.5th and 97.5th give 95%.

Marginalization. To get $p(\theta_1 | x)$, simply histogram the θ_1 values from your samples. Marginalization—usually a hard integral—becomes trivial.

Derived quantities. Want the posterior over some function $g(\theta)$? Evaluate g on each sample and histogram the results.

8 When MCMC doesn't scale

MCMC gives us exact samples from the posterior, but it can be slow. What about models with hundreds or thousands of parameters? What about applications requiring real-time inference?

Variational inference (VI) trades exactness for speed. Instead of sampling from the posterior, we approximate it with a simpler distribution. The result is an optimization problem rather than a sampling problem.

9 Approximation as optimization

Choose a family $\mathcal{Q}$ of tractable distributions (e.g., Gaussians) and find the member q^* that best approximates the true posterior. Once we have q^* , we use it in place of the posterior for all downstream tasks.

10 Measuring closeness: the KL divergence

The Kullback–Leibler divergence measures how much information is lost when we use q to approximate p :

$$\text{KL}(q\|p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta | x)} d\theta \quad (10.1)$$

We want $q^* = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\theta)\|p(\theta | x))$.

But there's an obstacle: computing the KL requires the normalizing constant $p(x)$, which is exactly the intractable integral we're trying to avoid.

11 The evidence lower bound (ELBO)

Rearranging the KL divergence:

$$\log p(x) = \underbrace{- \int q(\theta) \log \frac{q(\theta)}{\tilde{p}(\theta | x)} d\theta}_{\text{ELBO}(q)} + \text{KL}(q\|p) \quad (11.1)$$

Since $\text{KL} \geq 0$, we have $\text{ELBO}(q) \leq \log p(x)$ —the ELBO is a lower bound on the log-evidence.

Maximizing the ELBO is equivalent to minimizing the KL divergence. And the ELBO only involves the unnormalized posterior, which we can evaluate.

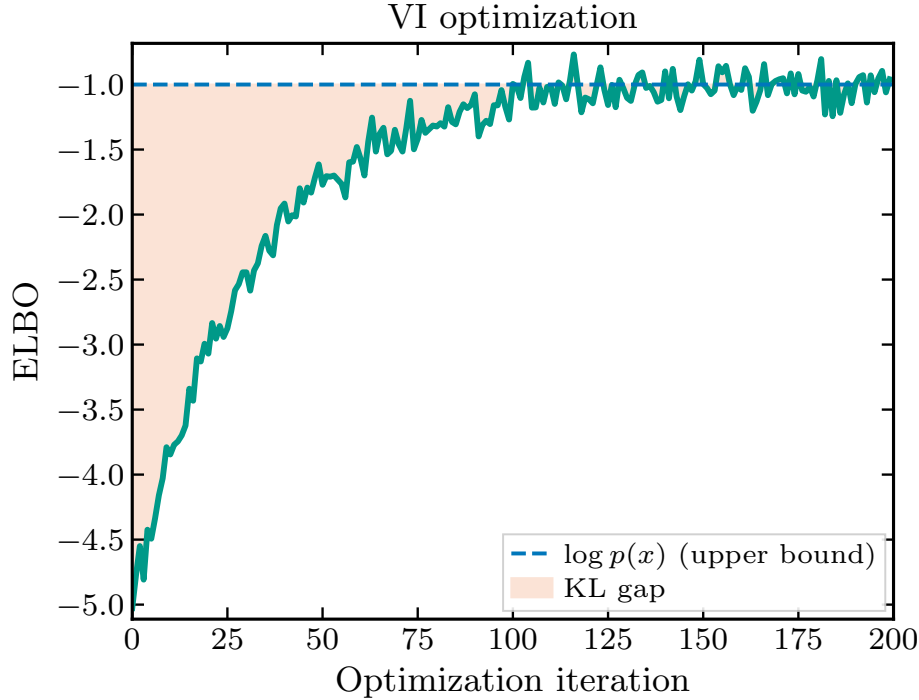


Figure 6. Variational inference as optimization. We search over a family of simple distributions (here, axis-aligned Gaussians) to find the one closest to the true posterior. The optimization adjusts the mean and variance parameters to maximize the ELBO.

12 The two forces in the ELBO

$$\text{ELBO}(q) = \underbrace{\mathbb{E}_q[\log p(x | \theta)]}_{\text{expected log-likelihood}} - \underbrace{\text{KL}(q(\theta) \| p(\theta))}_{\text{divergence from prior}} \quad (12.1)$$

Two opposing forces:

- The first term rewards q for concentrating on parameters that explain the data well.
- The second term penalizes q for deviating from the prior.

13 Variational families

The choice of $\mathcal{Q}$ controls the expressiveness–tractability tradeoff.

Mean-field Gaussian. $q(\theta) = \prod_j \mathcal{N}(\theta_j; \mu_j, \sigma_j^2)$. Each parameter is independent. Simple and fast, but cannot capture correlations.

Full-covariance Gaussian. $q(\theta) = \mathcal{N}(\theta; \mu, \Sigma)$. Captures correlations but requires $O(d^2)$ parameters.

Normalizing flows. Transform a simple base distribution through invertible maps. Can approximate complex, multimodal posteriors.

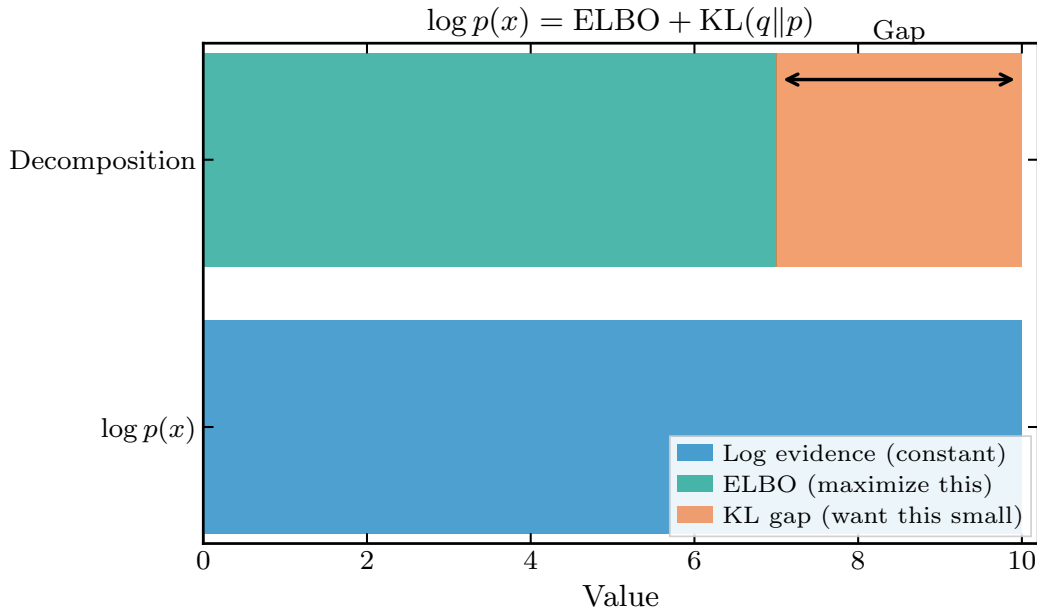


Figure 7. The two terms in the ELBO. The expected log-likelihood (left) rewards q for placing mass where the data are well-explained. The KL from prior (right) penalizes deviation from prior beliefs. The optimal q balances these forces.

14 Mode-seeking behavior

$\text{KL}(q||p)$ penalizes q for placing mass where p is small. The consequence: q is *mode-seeking*. It concentrates on one mode of the posterior and ignores others.

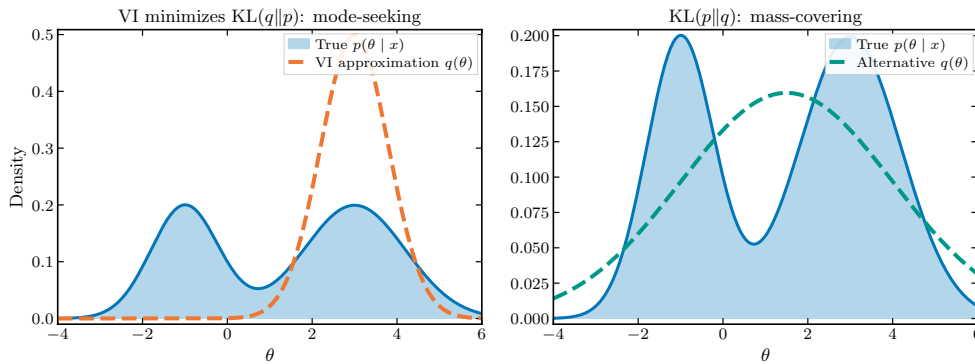


Figure 8. Mode-seeking behavior of VI. When the true posterior (gray) is multimodal, minimizing $\text{KL}(q||p)$ causes the variational approximation q (orange) to concentrate on a single mode rather than covering all modes. Placing q mass where p is near zero incurs a large penalty.

Practical consequence: underestimated uncertainty.— Because VI is mode-seeking, variational approximations typically underestimate posterior variance. Credible intervals may be too

narrow. If accurate uncertainty quantification is critical, MCMC may be the better choice despite its computational cost.

15 VI versus MCMC

	MCMC	Variational Inference
Output	Samples	Distributional approximation
Asymptotic accuracy	Exact	Limited by family $\mathcal{Q}$
Speed	Can be slow	Fast (optimization)
Multimodality	Can explore all modes	Typically mode-seeking
Uncertainty	Generally well-calibrated	Often underestimates

MCMC is the right choice when accurate uncertainty quantification matters. VI is the right choice when speed matters and some approximation error is acceptable.

Key Takeaways

- Monte Carlo replaces integrals with averages over samples; error is $O(1/\sqrt{N})$, independent of dimension.
- Metropolis–Hastings biases a random walk via an acceptance ratio. Normalizing constants cancel.
- Random walk Metropolis scales poorly ($\tau \propto d$); HMC uses gradients for near-dimension-independent τ .
- Autocorrelation time is the key diagnostic: it tells you how many independent samples you actually have.
- VI approximates the posterior by maximizing the ELBO, which is equivalent to minimizing KL divergence.
- The ELBO balances fitting the likelihood against staying close to the prior.
- VI is mode-seeking and typically underestimates uncertainty; MCMC is exact but slower.